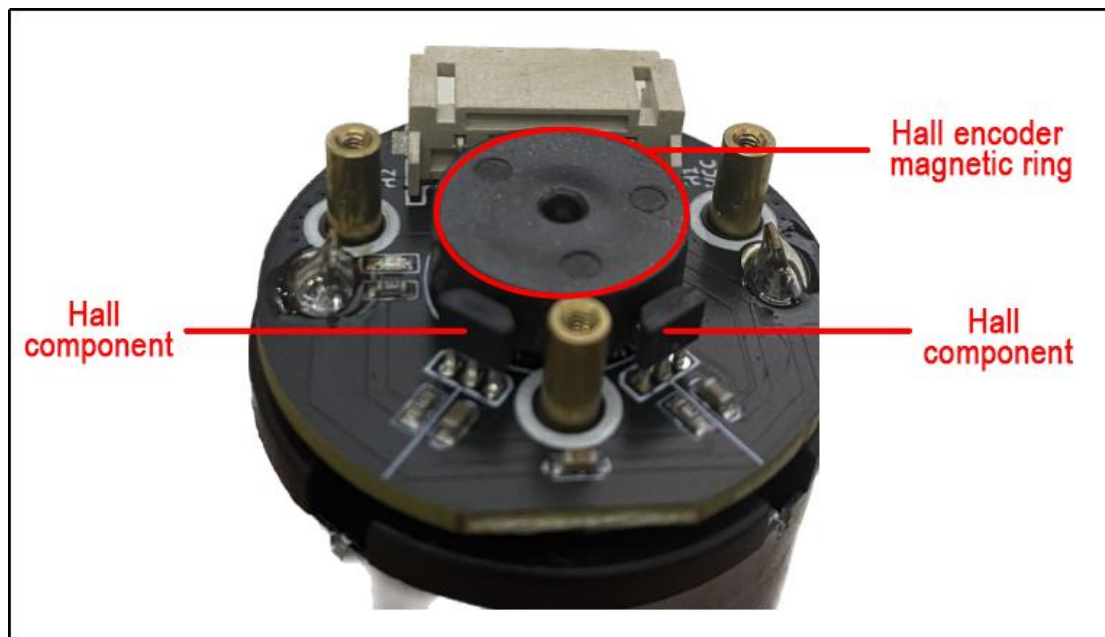
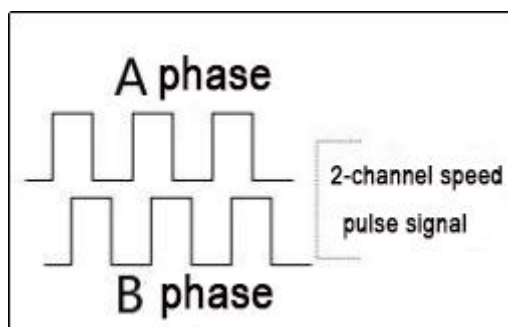


Encoder Usage Principle

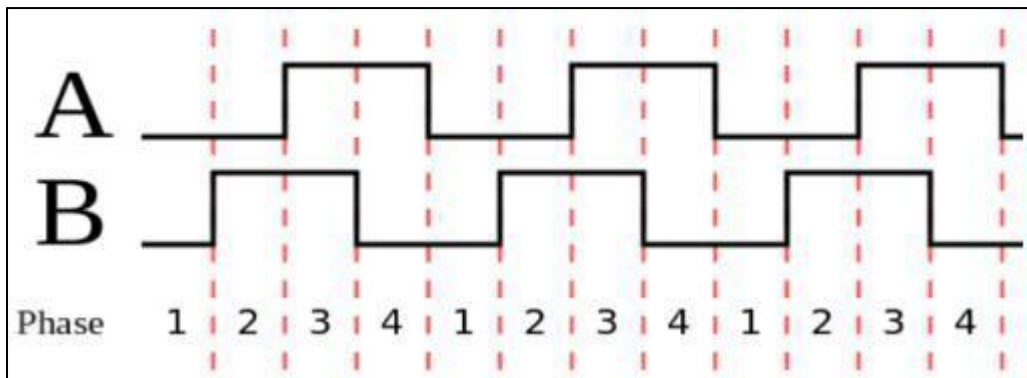
The Hall encoder disk is aligned with the motor's shaft. As the motor rotates, the Hall device detects and generates a series of pulse signals. To ascertain the steering direction, it typically produces two sets of square wave signals with a specific phase difference.



Then, the microcontroller can gather pulse data from the AB-phase. By converting these pulses, the motor's position can be determined.



To enhance encoder accuracy, we can employ a quadruple-frequency data processing method to analyze the data.



The presented waveform diagram is the output from the encoder. Typically, the standard processing approach involves using phase A for counting and phase B for determining the current rotation direction. Implementation examples include counting the rising or falling edge of phase A while simultaneously assessing the direction based on the level state of phase B during the said edges.

The quadruple frequency method involves calculating each transition edge of the AB two-phase simultaneously, achieving four counts within one pulse period of the A-phase count, thereby enhancing accuracy. STM32 provides its encoder interface, allowing direct use of the hardware interface (timer CH1&CH2) for counting, and reading corresponding data from the related registers. The STM32 program primarily initializes the timer into encoder mode.

For microcontrollers lacking an encoder hardware interface, external interrupt interfaces can be utilized. For instance, connecting the A phase output of the encoder to the microcontroller's external interrupt interface allows triggering interrupts at transition edges. In the corresponding interrupt service function, the forward and reverse directions can be determined by assessing the level of phase B. For example, during a transition edge in phase A, if phase B is at a high level, it indicates forward rotation; if at a low level, it signifies reverse rotation.